

KLH University · Sec 10 · Team 4

Interview Practice Platform

AI Resume Matching & Feedback

Database Systems Engineering and Distributed Backend Development (25CS1302E)

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Problem Statement

Job seekers struggle to find roles that genuinely fit their skills, and even when they land an interview, most prepare with generic questions that don't reflect the role — or get zero feedback on how they actually performed.

! **No personalized job matching**
Resumes are manually screened or keyword-filtered, missing better-fit roles a candidate never sees.

! **Generic interview prep**
One-size-fits-all question banks don't target the specific role's likely questions or a candidate's own gaps.

! **No feedback loop**
Practicing alone gives no insight into answer quality, pacing, or delivery — the things that actually decide interviews.

The Gap Today

75%+

of resumes are filtered out by ATS or recruiters before a human ever reads them

Few

candidates can access affordable, role-specific interview coaching

Zero

structured feedback on delivery when practicing alone at home

Objectives

What this project sets out to achieve, mapped to the problems identified above.

01

Accurate resume parsing

Extract skills, experience and role signals from a candidate's resume reliably.

02

Smarter job matching

Rank job fit using semantic similarity rather than keyword filters alone.

03

Role-specific mock interviews

Generate interview questions tailored to the matched job and the candidate's gaps.

04

Actionable feedback

Score spoken answers on content and delivery so candidates know what to improve.

05

Track progress over time

Store sessions, transcripts and scores so candidates can see improvement across attempts.

06

Keep it accessible

Build entirely on free-tier and open-source infrastructure so it stays affordable.

Literature Survey — Resume & Job Matching

Prior work this project builds on for parsing resumes and scoring candidate-job fit.

Wadhwa, Malhotra & Saxena (2020) *Automated Resume Screening System using NLP*

IJERT, Vol. 9, Issue 5

Early rule- and NLP-based pipeline for screening resumes against job requirements; established the parse-then-score pattern this project follows.

Khatrī et al. (2025) *AI-Powered Automated Resume Screening and Job Matching System Using NLP and ML*

IJRESM, Vol. 8, No. 10

Uses Sentence-BERT to embed resumes and job descriptions and rank fit by cosine similarity — the same embedding-based matching approach used by this platform's sentence-transformers layer.

JobMatchAI (2026) *An Intelligent Job Matching Platform Using Knowledge Graphs, Semantic Search & Explainable AI*

arXiv preprint

Combines multi-strategy resume parsing (regex + NER) with semantic skill matching, and reports the extraction accuracy trade-offs across name, skills and education fields.

Literature Survey — Interview Practice Tools

Existing mock-interview and speech-feedback tools, and the gap between them this project targets.

Existing Tools

Google Interview Warmup — Free browser tool that transcribed spoken answers to five questions and showed word-pattern feedback. Retired by Google in April 2026.

Pramp / Exponent Practice — Peer-to-peer mock interviews for coding and behavioral rounds — human feedback, not AI-scored, and metered by credits since its 2024 move to Exponent.

Yoodli — AI delivery coach scoring pace, filler words and confidence from speech — strong on delivery, but not tied to a candidate's resume or a specific job.

OpenAI Whisper (Radford et al., 2022) — Large-scale weakly-supervised speech recognition model — the open-source ASR backbone (via faster-whisper) this project uses for transcription.

The Gap This Project Fills

- Resume-matching research (previous slide) stops at ranking candidates — it doesn't lead into interview practice.
- Delivery coaches like Yoodli score speech but ask generic questions, not ones tied to a matched job or the candidate's resume gaps.
- Peer platforms like Pramp depend on finding a human partner and don't score content automatically.
- No surveyed tool combines resume-to-job matching, role-specific question generation, and content + delivery scoring in one free-tier pipeline — which is the integration this platform provides.

Proposed System

An AI-powered platform that matches candidates to roles from their resume, generates role-specific mock interview questions, captures spoken answers in-browser, and returns structured feedback — all running on free-tier and open-source infrastructure.



Resume & Matching Service

- spaCy — resume parsing, skill/experience extraction
- sentence-transformers — embedding-based job similarity
- Seeded job dataset for match ranking
- REST endpoints consumed by the Node backend

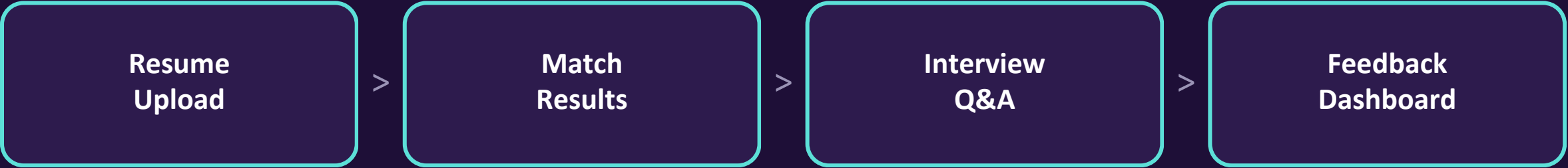


Interview & Feedback Engine

- getUserMedia + MediaRecorder — in-browser mic/camera capture
- faster-whisper — speech-to-text transcription
- Groq/Gemini LLM — question generation & feedback
- MongoDB — sessions, transcripts & scores

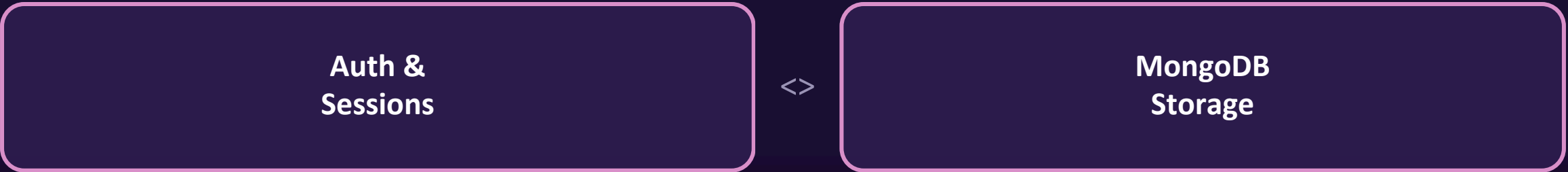
System Architecture

Frontend (React)



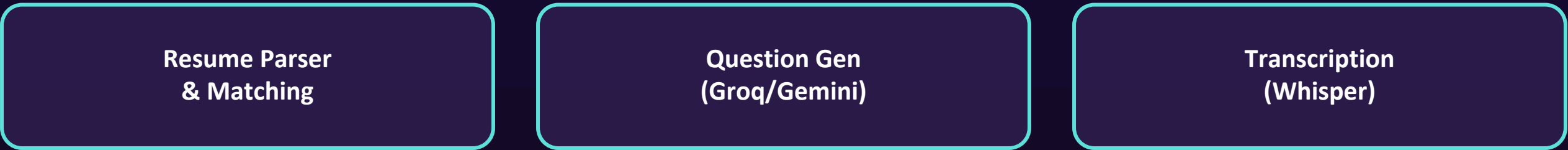
↓ *REST calls, auth via Node backend*

Backend (Node.js + MongoDB)



↓ *Orchestrates calls to the AI microservice*

AI Service (Python / Flask)



Design Notes

- Node and Python are separate services talking over REST — AI logic stays isolated and swappable.
- All AI components run on free-tier / open-source tools: Groq, faster-whisper, sentence-transformers.
- MongoDB's flexible schema suits variable-length transcripts and evolving question sets.
- Stack: React.js, Node.js + Express, Python (Flask), spaCy, sentence-transformers, faster-whisper, Groq/Gemini API, MongoDB Atlas, Git/GitHub.

Modules / Functionalities

01

Resume Upload & Parsing

Extracts skills and experience from resume text using NLP-based parsing.

02

Job Matching Engine

Scores candidate-job fit using embedding-based similarity against a job dataset.

03

AI Question Generation

Generates role-specific mock interview questions from the matched job and resume gaps.

04

Live Answer Capture

In-browser mic and camera recording — no plugins or external apps required.

05

Speech-to-Text & Analysis

Transcribes spoken answers and surfaces pacing and filler-word signals.

06

AI Feedback & Scoring

LLM-based feedback on content and delivery, stored per session for progress tracking.

Frontend Design

Key screens in the candidate-facing React application.

Home Page

Landing page introducing the platform and entry point to upload a resume.

Login / Registration

Candidate authentication — sign up, log in, and session handling.

Forms

Resume upload and job-preference forms feeding the matching engine.

Dashboard

Match results, mock interview access and feedback history in one place.

Frontend Implementation — Screenshots

Interview Practice

Sign in to continue your prep.

Email

you@example.com

Password

.....

→] Log in

or

Continue as guest

This is a demo login — any email/password works.

Role-Based Practice

Upload your resume

We'll match you to roles based on your skills and experience.

Drag and drop your resume, or click to browse

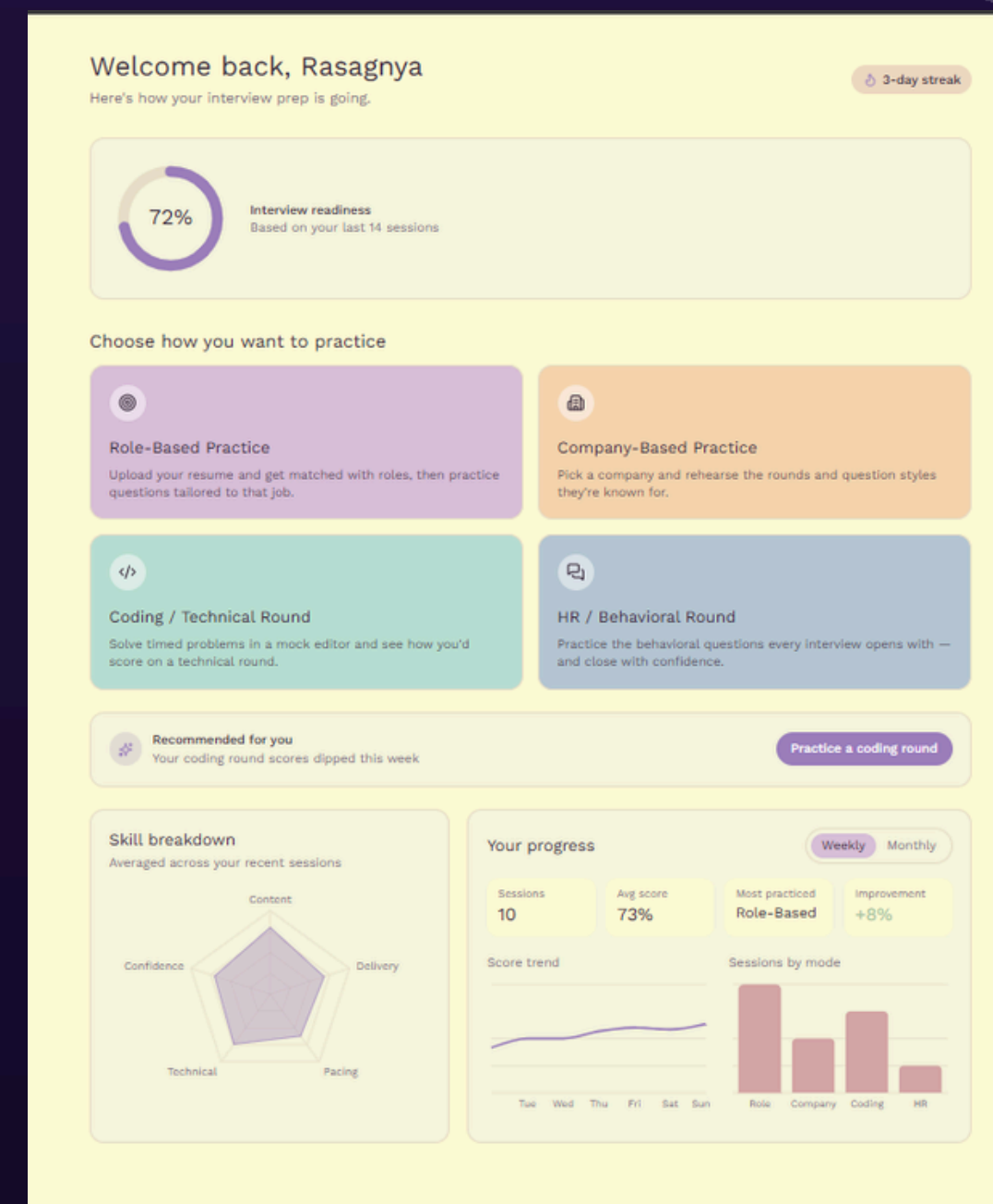
PDF or Word, up to 5MB

or

Paste your resume text instead

Find matching roles

[Use sample resume](#) · [Download it to upload manually](#)

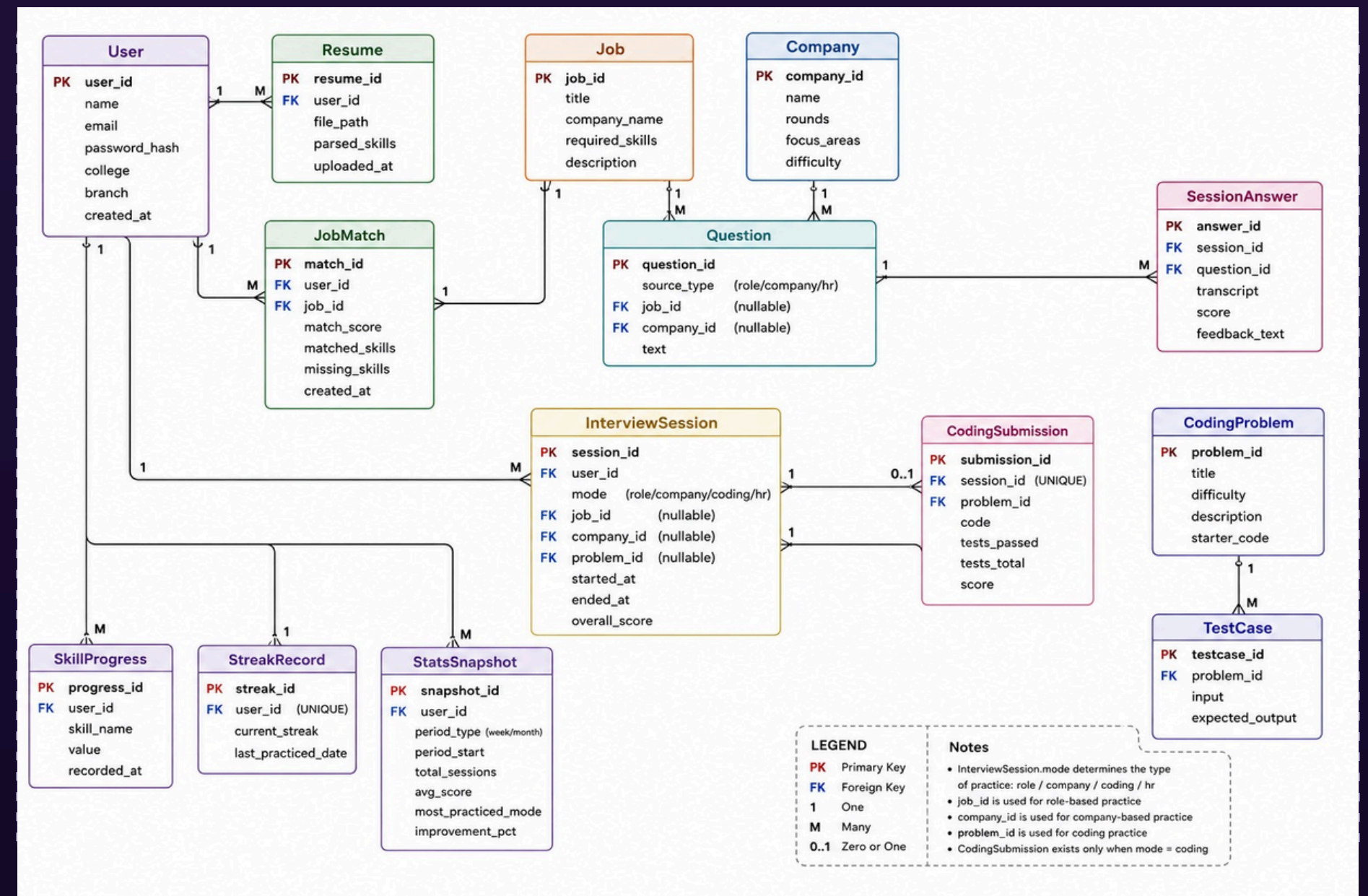


Database Design & ER Diagram

MongoDB collections for candidates, resumes, jobs, sessions and feedback, and how they relate.

Design Principles

- Flexible, document-based schema to suit variable-length transcripts
- Collections: Users, Resumes, Jobs, InterviewSessions, Feedback
- References between collections via ObjectId rather than deep nesting
- Indexes on candidate ID and session ID for fast dashboard lookups



Database Tables / Schema & Sample Records

Schema Overview

Users name, email, passwordHash, createdAt

Resumes userId, parsedSkills[], rawText, uploadedAt

Jobs title, description, requiredSkills[], embedding

InterviewSessions userId, jobId, questions[], transcript, status

Feedback sessionId, contentScore, deliveryScore, comments

InterviewIQ — Sample Database Records

Users

user_id	name	college	branch
1	Niyathi	KL University	IT
2	Rahul Sharma	ABC Engineering	CSE

Jobs

job_id	title	company_name
1	Data Analyst	Microsoft
2	Software Engineer	Google
3	BI Analyst	Amazon

InterviewSessions

session_id	user_id	mode	score
1	1	role	82.50
2	1	coding	90.00

Questions

question_id	source_type	question_text
1	role	Explain INNER JOIN vs
2	role	What is data normaliza
3	company	Why do you want to wo
4	hr	Tell me about yourself.

Sample records from the InterviewIQ MySQL database

Current Progress

Resume parsing service

[In progress]

Job matching engine

[In progress]

Question generation (LLM)

[Not started]

Live answer capture (audio/video)

[Not started]

Speech-to-text pipeline

[Not started]

Feedback dashboard (frontend)

[Not started]

Update the status labels above with the team's actual progress before Review 2.

Challenges / Issues Faced

Open technical questions that come with this three-service, free-tier architecture.



Inconsistent resume formats

spaCy-based parsing works well on clean, single-column resumes but accuracy drops on tables, multi-column layouts and graphic-heavy templates — pushing us toward extra pre-processing and fallback rules rather than parsing alone.



Latency across three services

A single mock-interview turn chains Node → Flask → an LLM call (and Whisper for transcription). Keeping that feel responsive in a live, in-browser session means careful async handling, not just faster individual calls.



Free-tier rate limits

Groq/Gemini and MongoDB Atlas free tiers cap request volume and concurrency, so testing with multiple simultaneous mock sessions needs request queuing and graceful backoff rather than firing calls directly.



Cross-browser audio/video capture

getUserMedia and MediaRecorder support different codecs and permission flows across browsers, so a recording flow that works in one browser needs explicit fallback handling to stay consistent in others.

Future Work

● Body Language Analysis

MediaPipe-based heuristics for eye contact, posture and fidgeting during mock interviews.

● Voice Modulation Scoring

librosa-based pitch, pace and volume analysis for a fuller picture of delivery.

● Recruiter Dashboard

A recruiter-facing view into candidate readiness and match pipelines.

● Live Job Board Integration

Pull real postings via LinkedIn/Indeed APIs instead of a static seed dataset.

● Multi-Language Interviews

Support mock interviews and feedback in regional languages.

● Peer Mock Scheduling

Let candidates practice live with peers, with AI scoring both sides.

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